

Abstract algebra – 7

Generators and relations: infinite and matrix groups

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Theoretical minimum. An infinite group is usually given by generators, and it is known only when we know which words in them are trivial. To identify a group from a list of relations one bounds the group they define from above by a normal form and realises the bound by a model — matrices or geometric transformations; to show that there are no relations at all one lets the group act and plays ping-pong.

Definitions. The free group $F(S)$ consists of the reduced words in the letters $s^{\pm 1}$, $s \in S$ (no ss^{-1} or $s^{-1}s$ inside), multiplied by concatenation and cancellation; $F_2 = F(a, b)$. For a set R of words, $\langle S \mid R \rangle = F(S)/N$, where N is the smallest normal subgroup containing R (a presentation). The free product $G * H$ is presented by the generators and relations of both factors, e.g. $\mathbb{Z}_3 * \mathbb{Z}_3 = \langle x, y \mid x^3 = y^3 = e \rangle$. A group is finitely generated if $G = \langle g_1, \dots, g_k \rangle$; it is torsion if all its elements have finite order, and of exponent m if $g^m = e$ for all g . An involution is an element with $s^2 = e$. The Coxeter group of a symmetric matrix (m_{ij}) , where $m_{ii} = 1$ and $2 \leq m_{ij} \leq \infty$ for $i \neq j$, is $\langle s_1, \dots, s_k \mid (s_i s_j)^{m_{ij}} = e \rangle$ (no relation where $m_{ij} = \infty$).

Theorem 1 (von Dyck, Moore). If elements g_s of a group G satisfy the relations R , then $s \mapsto g_s$ extends to a homomorphism $\langle S \mid R \rangle \rightarrow G$, onto if the g_s generate G . So to prove $G \cong \langle S \mid R \rangle$ one checks the relations in G and shows that $\langle S \mid R \rangle$ has at most $|G|$ elements. The reflections s, t of a regular n -gon in two axes at angle π/n give $D_n \cong \langle s, t \mid s^2 = t^2 = (st)^n = e \rangle$, since every word reduces to $(st)^k$ or $(st)^k s$; the maps $x \mapsto -x$, $x \mapsto 1 - x$ of $\mathbb{Z}$ give the infinite dihedral group $\mathbb{Z}_2 * \mathbb{Z}_2 = \langle s, t \mid s^2 = t^2 = e \rangle$. (Moore) With $s_i = (i \ i+1)$,

$$S_n \cong \langle s_1, \dots, s_{n-1} \mid s_i^2 = (s_i s_{i+1})^3 = (s_i s_j)^2 = e \text{ for } |i - j| \geq 2 \rangle.$$

Idea. In the group defined by the relations, $H = \langle s_1, \dots, s_{n-2} \rangle$ has at most $(n-1)!$ elements by induction, and the n cosets $Hs_{n-1}s_{n-2} \cdots s_k$ ($1 \leq k \leq n$)

are permuted by right multiplication by every s_i , so there are no others. Generation of S_n itself is Abstract algebra – 3.

Fact 1 (finite generation). Quotients and subgroups of finite index of finitely generated groups are finitely generated (Schreier; a subgroup of index m in F_k is free of rank $m(k-1)+1$), and so are subgroups of finitely generated abelian groups (Linear algebra – 9); arbitrary subgroups need not be. To prove that G is *not* finitely generated, write it as a strictly increasing union of subgroups (like $\mathbb{Q} = \bigcup_n \frac{1}{n!}\mathbb{Z}$), map it onto such a group, or find a quantity (a denominator, a degree) bounded on each finite set of words.

Theorem 2 (ping-pong). Let a, b act on a set X and let $X_1, X_2 \subseteq X$ be disjoint and non-empty with $a^k(X_2) \subseteq X_1$ and $b^k(X_1) \subseteq X_2$ for all $k \neq 0$. Then $\langle a, b \rangle \cong F_2$. *Proof.* A non-trivial reduced word is conjugate to some $w = a^{k_1}b^{l_1} \dots b^{l_{r-1}}a^{k_r}$ with non-zero exponents; reading from the right, $w(X_2) \subseteq X_1$, so $w \neq e$. *Example* (Sanov): $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ act on $\mathbb{R}^2$ with $X_1 = \{|x| > |y|\}$, $X_2 = \{|x| < |y|\}$, because $|x + 2ky| > |y|$ when $|x| < |y|$ and $k \neq 0$. Together with $-I$ they generate the kernel of $\mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}_2)$, so $\mathrm{SL}_2(\mathbb{Z})$ has a free subgroup of index 12 (statement only).

Fact 2 (involutions and reflections). g is a product of two involutions if and only if $sgs^{-1} = g^{-1}$ for some involution s (put $t = sg$). The reflections in two lines meeting at angle θ compose to the rotation by 2θ ; so every element of D_n and of $O(2)$ is a product of two reflections, and reflections in mirrors at angles π/m_{ij} satisfy the Coxeter relations. Every bijection of a set is a product of two involutions (Combinatorics – 8, homework 2).

Theorem 3 (Minkowski). Let p be an odd prime. If $A \in \mathrm{GL}_n(\mathbb{Z})$ has finite order and $A \equiv I \pmod{p}$, then $A = I$. *Proof.* Otherwise some power of A has prime order q ; write it as $I + p^k B$ with $k \geq 1$, $B \not\equiv 0 \pmod{p}$, and expand $(I + p^k B)^q = I$ modulo p^{k+1} if $q \neq p$, modulo p^{k+2} if $q = p$ (here $p \geq 3$ is used); both give $B \equiv 0 \pmod{p}$. Hence reduction modulo p embeds every finite subgroup of $\mathrm{GL}_n(\mathbb{Z})$ into $\mathrm{GL}_n(\mathbb{F}_p)$. For $p = 2$ the claim fails: $-I \equiv I$.

Theorem 4 (Burnside). A subgroup $G \subseteq \mathrm{GL}_n(\mathbb{C})$ of finite exponent m is finite, $|G| \leq m^{n^3}$. *Idea.* If G acts irreducibly on $\mathbb{C}^n$, it spans $M_n(\mathbb{C})$ (Burnside's theorem on matrix algebras; quote it). For a basis $g_1, \dots, g_{n^2} \in G$ the map $g \mapsto (\mathrm{tr} gg_i)_i$ is injective, the trace form being non-degenerate,

and each $\text{tr } gg_i$ is a sum of n roots of unity of degree m (Abstract algebra – 5). If G is reducible, make it block triangular and induct: the kernel of the map to the diagonal blocks consists of unipotent matrices of finite order, which are trivial in characteristic 0. (Schur: a finitely generated torsion subgroup of $\text{GL}_n(\mathbb{C})$ is finite.)

1. Prove Theorem 1: the presentations of D_n ($n \geq 3$), of the infinite dihedral group (show that the words $(st)^k$ and $(st)^k s$, $k \in \mathbb{Z}$, give pairwise different maps of $\mathbb{Z}$) and Moore's presentation of S_n . Deduce that in a group the product of two elements of order 2 may have infinite order, so the elements of finite order need not form a subgroup.
2. Prove the ping-pong lemma and deduce that for every real $\lambda \geq 2$ the matrices $\begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}$ generate a free group of rank 2. Show that for $\lambda = 1$ they do not, by finding a non-trivial reduced word in them that is equal to a matrix of finite order.
3. Let G be the subgroup of $\text{GL}_2(\mathbb{R})$ generated by $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Let H consist of those matrices $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ in G for which $a_{11} = a_{22} = 1$. (a) Show that H is an abelian subgroup of G . (b) Show that H is not finitely generated. (IMC 1996)
4. (a) Prove that an element g of a group is a product of two involutions if and only if $sgs^{-1} = g^{-1}$ for some involution s . (b) Prove that $A \in \text{GL}_n(\mathbb{C})$ is a product of two involutions of $\text{GL}_n(\mathbb{C})$ if and only if A is similar to A^{-1} . *Hint:* for a Jordan block with eigenvalue 1 consider the operators $p(x) \mapsto p(x+1)$ and $p(x) \mapsto p(-x)$ on the polynomials of degree less than n .
5. Let $F = A_0 A_1 \dots A_n$ be a convex polygon in the plane. Define for all $1 \leq k \leq n-1$ the operation f_k which replaces F with a new polygon $f_k(F) = A_0 \dots A_{k-1} A'_k A_{k+1} \dots A_n$, where A'_k is the point symmetric to A_k with respect to the perpendicular bisector of $A_{k-1} A_{k+1}$. Prove that $(f_1 \circ f_2 \circ \dots \circ f_{n-1})^n(F) = F$. We suppose that all operations are well-defined on the polygons to which they are applied, i.e. the results are convex polygons again. ($A_0, A_1, \dots, A_n$ are the vertices of F in consecutive order.) (IMC 2011)